Question	Answer	Marks	Partial Marks
7(b)	76.5 or 76.52 to 76.53	**3**	**M2** for $[\sin =]\frac{16.7\sin 32}{9.1}$ oe or **M1** for $\frac{9.1}{\sin 32}=\frac{16.7}{\sin M}$ oe
7(c)(i)	$\frac{1}{2}\times 12.3\times 21.5\sin(...)=62.89$ or better	**M1**	
	28.40 to 28.41…	**A1**	
7(c)(ii)	12.2 or 12.17 to 12.18	**3**	**M2** for $\sqrt{12.3^2+21.5^2-2\times 12.3\times 21.5\times\cos 28.4}$ OR **M1** for $12.3^2 + 21.5^2 – 2 \times 12.3 \times 21.5 \times \cos28.4$ **A1** for 148 or 148.2 to 148.3
7(c)(iii)	6.6[0] to 6.62	**3**	**M2** for 21.5cos28.4 – 12.3 or **M1** for 21.5cos28.4
8(a)(i)	$\frac{1}{6}$ oe	**1**	
8(a)(ii)	25	**1**	**FT** *their* **(a)(i)** dep on 0 < **(a)** < 1
8(b)(i)	$\frac{11}{36}$ oe	**3**	**M2** for $\frac{1}{6}\times\frac{2}{6}+\frac{3}{6}\times\frac{3}{6}$ oe or correct possibility diagram with 11 outcomes identified or **M1** for $\frac{1}{6}\times\frac{2}{6}$ or $\frac{3}{6}\times\frac{3}{6}$ oe or lists the 11 required outcomes or for possibility diagram but required outcomes not indicated
8(b)(ii)	$\frac{2}{11}$ oe	**2**	**M1** for $\frac{2}{k}$ or $\frac{p}{their\,11}$ seen oe leading to answer
8(c)	6	**2**	**M1** for $\left(\frac{4}{6}\right)^k\times\frac{2}{6}=\frac{32}{729}$ written oe soi by one trial with $k > 1$ or $2^{n-1}=32$ or better or $3^n=729$ or better
9(a)	$12x^2-4x^3$ oe final answer	**2**	**B1** for $12x^2$ or $-4x^3$ in final answer or for correct answer seen

Question	Answer	Marks	Partial Marks
9(b)	(3, 27)	**3**	**B2** for $x = 3$ OR **M1** for *their* $12x^2 - 4x^3 = 0$ or better or states $\frac{dy}{dx} = 0$ **M1dep** for substituting *their x* into $y = 4x^3 - x^4$ shown
9(c)	–64	**3**	**M1** for $4x^3 - x^4 = 0$ **B1** for $x = 4$
10(a)	[*DEF*], *BCD* *ADF*, *ADB*	**2**	**B1** for each pair
10(b)	*OQ* *OQT* Tangent perpendicular to radius RHS equal	**5**	**B1** for each
11(a)	4	**1**	
11(b)	$7 - 3x$ final answer	**2**	**M1** for $1 - 3(x - 2)$
11(c)	$\frac{1-x}{3}$ oe final answer	**2**	**M1** for $x = 1 - 3y$ or $y - 1 = -3x$ or $1 - y = 3x$ or $\frac{y}{3} = \frac{1}{3} - x$
11(d)	$a = 2$, $b = 5$, $c = -1$	**5**	**B4** for two correct values <u>only</u> after correct substitution seen i.e. $(1 - 3x - 1)^2 - (x - 1)^2(1 - 3x)$ or for correct unsimplified expansion or a correct simplified expansion. OR **M1** for $(1 - 3x - 1)^2 - (x - 1)^2(1 - 3x)$ **B2** for correct expansion of $[-](x - 1)^2(1 - 3x)$ $[-](x^2 - x - x + 1 - 3x^3 + 3x^2 + 3x^2 - 3x)$ or better or **B1** for expansion of one pair of brackets $\left[(x-1)^2 = \right]x^2 - x - x + 1$ or better or $[(x - 1)(1 - 3x) =] - 3x^2 + x + 3x - 1$